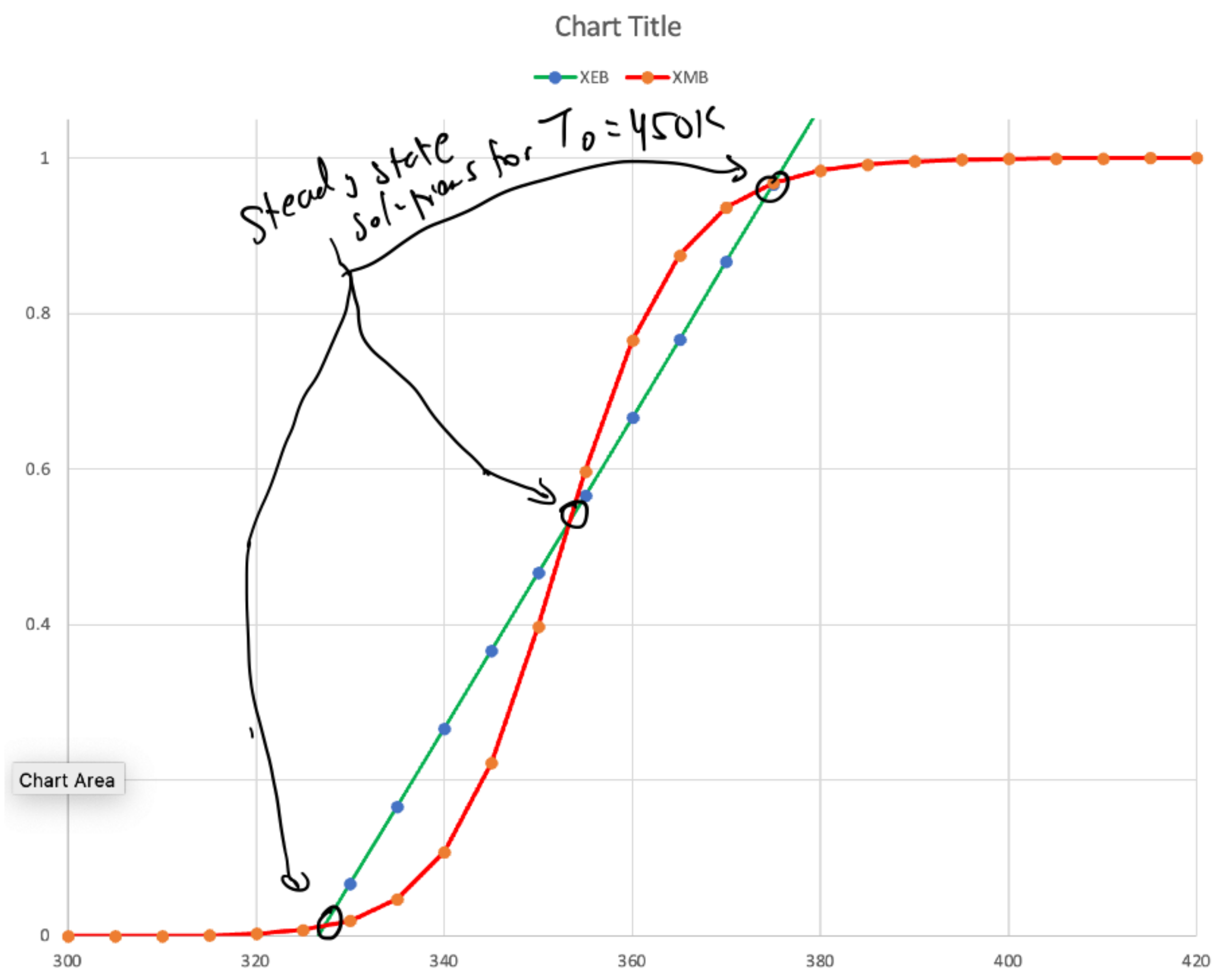
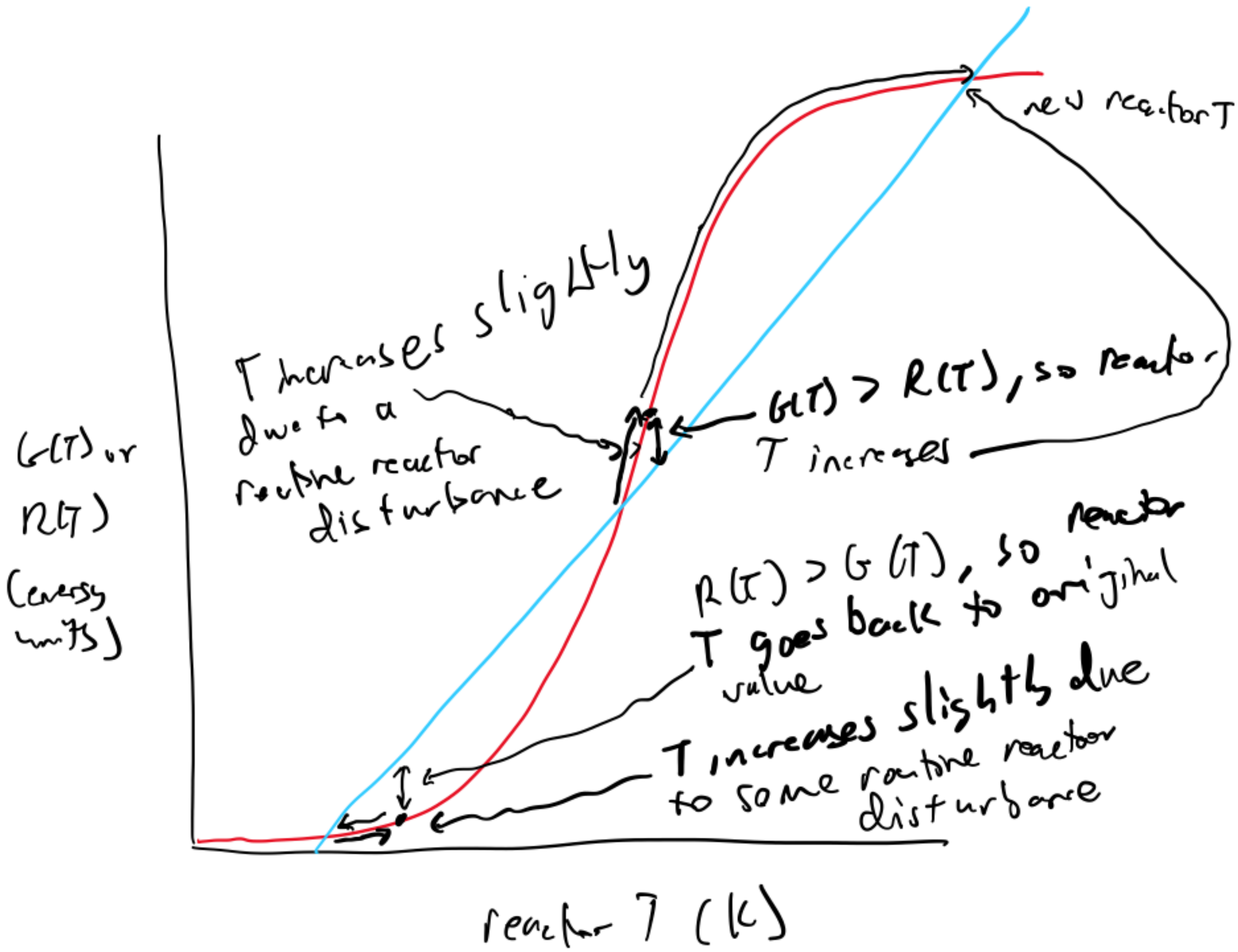
18 Steady state stability



- The steady state temperatures are the temperatures where $X_{EB} = X_{MB}$, i.e., where the X_{EB} & X_{MB} curves intersect
- The temperatures on the x-axis are
  reactor temperatures
- There can be multiple points of intersection for one feed temperature. For example, the feed temperature above has 3 steady state solutions.

- Steady state stability depends on $G(T)$ & $R(T)$ for T just slightly greater than that at the intersection point
- If G(T) > R(T) the solution is unstable. This is b/c if the reactor temperature increases slightly, the heat exchanger can't quench !
- If R(T) > G(T) the solution is stable, since the small increase in T will bring the reactor temp back down to this steady state.
- Hence, in the plot above:
  - Reactor temperature = 324 K: $R(T) > G(T)$ for T just greater than 324 K, so this ss is stable
  - Reactor temperature = 354 K:
    $G(T) > R(T)$ for T just above 354 K, so this ss is unstable
  - Reactor temperature = 380 K:

$R(T) > G(T)$ for T just greater than 380 K, so this ss is stable


new reactor T
T increases slightly due to a routine reactor disturbance
G(T) > R(T), so reactor T increases
G(T) or R(T) (energy units)
R(T) > G(T), so reactor T goes back to original value
T increases slightly due to some routine reactor disturbance
reactor T (K)